Toy schematic: clustering labels are conditional; frozen dCST assignments are fixed

This schematic illustrates label definition only; it is not a benchmark of clustering performance.

A. Clustering-defined CST labels

Target samples	s1	s2	s3	s4	s5	s6
Cluster, k = 2	C1	C1	C1	C2	C2	C2
Cluster, k = 3	C1	C1	C2	C2	C3	C3
Add samples	C1	C2	C2	C3	C3	C3

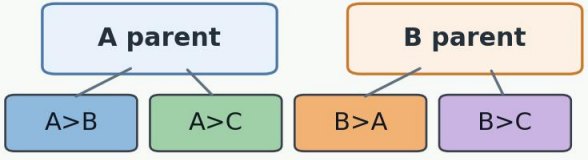
Interpretation

The label is conditional on choices such as k, distance, algorithm, initialization, and the comparison set used to fit the partition.

Same target samples; partitions can differ.

B. Frozen dCST assignment

Frozen reference hierarchy



Target samples	s1	s2	s3	s4	s5	s6
Before adding	A>B	A>B	A>C	B>A	B>C	A>C
After adding	A>B	A>B	A>C	B>A	B>C	A>C

Interpretation

Frozen hierarchy: aligned samples map to the same realized lineages.